From Ambiguity to Meaning: ArabicSeek's Semantic Search and Hybrid Summarization Pipeline

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***Abstract*—The growth of Arabic digital news has created a strong demand for retrieval systems that capture semantic intent rather than surface lexical overlap. Arabic compounds this challenge through rich morphology, dialectal and orthographic variation, and a severe query–document vocabulary mismatch. We present *ArabicSeek*, a fully implemented, end-to-end platform that addresses these issues with a twelve-stage pipeline combining dense neural retrieval, unsupervised topic discovery, and a faithfulness-first hybrid summarization module. The system embeds 8,635 articles from Al Jazeera's May 2026 Arabic archive with `intfloat/multilingual-e5-large` into 1,024-dimensional L2-normalized vectors and deploys a 3,480-article News search index, computing relevance as a single batched cosine-similarity operation with a mean latency of $0.19\,\mathrm{s}$ on a CPU instance without GPU acceleration. Topic structure is recovered by HDBSCAN [15] over UMAP [16] projections, yielding 20 coherent clusters labelled via class-based TF-IDF. For summarization, *ArabicSeek* deploys a faithful extractive summarizer (TextRank with MMR over E5 sentence embeddings) by default—chosen after comparing seven strategies that exposed the hallucination tendencies of small Arabic abstractive models—with an optional LLM refinement available on demand. Over 30 Arabic queries spanning five domains, *ArabicSeek* attains a mean NDCG@5 of 0.79 versus 0.61 for BM25, a statistically significant gain (paired Wilcoxon signed-rank test, $p < 0.01$), with the largest improvements on paraphrastic queries. The Streamlit front-end mirrors Al Jazeera's editorial design and fully supports right-to-left rendering alongside a 3D topic-galaxy visualization.**

***Index Terms*—Arabic NLP, semantic search, dense retrieval, multilingual embeddings, E5, BERTopic, HDBSCAN, extractive summarization, faithfulness, information retrieval**

I. Introduction

A. Motivation and Problem Statement

Arabic-language digital news has grown into one of the world's largest bodies of online text. Outlets such as Al Jazeera, BBC Arabic, and Al Arabiya collectively publish thousands of articles each day, accumulating archives that are rich in information yet increasingly difficult to navigate with conventional tools. Keyword-based search engines, which rank documents using term-frequency statistics such as BM25 [1], exhibit three systematic weaknesses when applied to Arabic. First, Arabic is morphologically among the most complex of the world's major languages: a single semantic root generates dozens of distinct surface forms through affixation and internal vowel change [2], so a query term may never appear verbatim in a semantically relevant document. Second, significant dialectal and orthographic variation means that the same concept may be rendered with different spellings across regional editions or individual authors. Third, users tend to formulate queries in natural-language prose that shares few or no tokens with relevant documents—the vocabulary mismatch problem [3]—a phenomenon that is more acute in Arabic than in English due to the denser morphological surface form. Collectively, these factors produce poor recall, missed synonyms, and a frustrating retrieval experience that erodes trust in search tools over long archives.

B. Neural Dense Retrieval as a Solution

Dense retrieval has emerged as the principled solution to vocabulary mismatch. The bi-encoder paradigm, established by DPR [4] and Sentence-BERT [5], encodes queries and documents into a shared vector space where geometric proximity approximates semantic similarity, decoupling retrieval quality from surface lexical overlap. The E5 family [6] extends this paradigm with a weakly supervised contrastive objective over web-scale multilingual text pairs, yielding representations that transfer to over 100 languages—including Arabic—without language-specific fine-tuning. Despite these advances, production-ready deployments for Arabic news remain scarce in the literature. The engineering challenges specific to Arabic—encoding heterogeneity, mixed numeral systems, right-to-left rendering requirements, and the absence of reliable Arabic sentence boundary detectors—have received limited systematic treatment. *ArabicSeek* addresses this gap with a complete, engineering-grade implementation.

C. Summarization Challenges for Arabic

Parallel progress in neural summarization has been partly impeded for Arabic by a shortage of large-scale, domain-matched training data. The mT5 model [8], trained on the multilingual mC4 corpus, provides a practical zero-shot baseline but is susceptible to hallucination on out-of-domain Arabic news text. AraBART [11] and AraT5 [12] address this through targeted Arabic pretraining, substantially improving factual consistency and fluency. *ArabicSeek* adopts a faithfulness-first hybrid strategy—deploying a faithful extractive summarizer by default while retaining abstractive options—and reports a comparative study of seven summarization methods in Section IX.

D. Contributions

This paper makes the following contributions:

1) **A complete twelve-stage NLP pipeline** from raw Arabic web scraping through evaluation, designed to be fault-tolerant and reproducible (§III).
2) **An ambiguity-aware preprocessing module** producing three specialized article views optimized for distinct downstream consumers (§IV).
3) **A robust multi-strategy corpus loader** with automatic encoding detection that handles real-world Arabic web data (§IV).
4) **A scalable offline pipeline** embedding 8,635 articles with `intfloat/multilingual-e5-large`, deploying a 3,480-article News search index, and discovering 20 coherent topic clusters over it via HDBSCAN on UMAP projections (§V).
5) **A CPU-deployable semantic search engine** with mean latency 0.19 s via exact cosine similarity and topic metadata enrichment (§VI).
6) **A faithfulness-first hybrid summarization module** pairing a TextRank+MMR extractive summarizer (deployed default) with an optional LLM refinement and a pattern-based key-point extractor, selected through a systematic comparison of seven strategies (§VII).
7) **An editorial-grade bilingual interface** with full RTL Arabic rendering, a 3D topic galaxy visualization, and a real-time breaking-news ticker (§VIII).

II. Related Work

A. Arabic Information Retrieval

Early Arabic IR systems applied root extraction or light stemming—notably the Khoja stemmer [17]—before applying BM25 or vector-space models. Empirical comparison showed that light stemming frequently outperforms aggressive root extraction because over-stemming collapses morphologically distinct but semantically different forms [18]. Contextual encoder models such as AraBERT [19] and CAMeL-BERT [20] achieve state-of-the-art results on Arabic NLU benchmarks but require bi-encoder fine-tuning to scale as retrieval systems. *ArabicSeek* avoids this dependency by exploiting the zero-shot multilingual retrieval capability of `intfloat/multilingual-e5-large`.

B. Multilingual and Cross-Lingual Retrieval

The LaBSE model [7] aligns embedding spaces across 109 languages through bilingual sentence pair pretraining. The E5 family [6] extends contrastive training to weakly supervised web-scale data. Neither model has been systematically evaluated for Arabic news retrieval in a production deployment; our work provides the first end-to-end empirical assessment in this setting. FAISS [21] and Haystack [22] provide infrastructure for large-scale deployments but introduce overhead unnecessary at corpus sizes below approximately 100,000 documents; we demonstrate that exact cosine similarity via NumPy is practical and faster for our 3,480-document search index.

C. Arabic Text Summarization

Extractive Arabic summarization has been approached through sentence selection [23] and minimum-redundancy maximum-relevance criteria [24]. LexRank [25] and SummaRuNNer [26] have been adapted to Arabic with varying success. AraBART [11] and AraT5 [12] provided the first Arabic-specific generative architectures with competitive benchmark performance. After systematically comparing seven strategies (Section IX), *ArabicSeek* deploys a faithful extractive summarizer as the default—prioritizing factual grounding over fluency—with AraT5 and an optional LLM available as abstractive alternatives; to our knowledge this is the most detailed such comparison reported for Arabic news summarization to date.

D. Unsupervised Topic Modeling

Latent Dirichlet Allocation [13] remains widely used but requires specifying the number of topics and assumes bag-of-words representations poorly suited to dense embeddings. BERTopic [14] combines UMAP dimensionality reduction with HDBSCAN clustering and class-based TF-IDF to produce interpretable, parameter-lean topic representations that operate directly on dense embeddings, making it particularly attractive for Arabic news where topic boundaries are often fuzzy.

III. System Architecture

ArabicSeek is organized into twelve discrete pipeline stages illustrated in Fig. 1. The pipeline partitions into four functional layers: (1) data acquisition, cleaning, and preprocessing; (2) linguistic feature extraction and E5 embedding; (3) clustering and search; and (4) summarization, evaluation, and interface. Each layer communicates via versioned flat files (JSONL, NPY, CSV), ensuring that stages can be re-run independently when parameters change.

All components are implemented in Python 3.10 using PyTorch, Hugging Face Transformers [27], Sentence-Transformers [5], CAMeL-Tools [20], and Streamlit [28].

IV. Data Acquisition and Preprocessing

A. Corpus Collection (Stage 0)

A total of 16,191 articles were scraped from `aljazeera.net` via a Scrapy-based sitemap and RSS collector. Each record carries five fields: `url`, `title`, `body`, `category`, and `date_published`. Duplicate detection uses SimHash over normalized body text, yielding 8,635 unique articles after de-duplication.

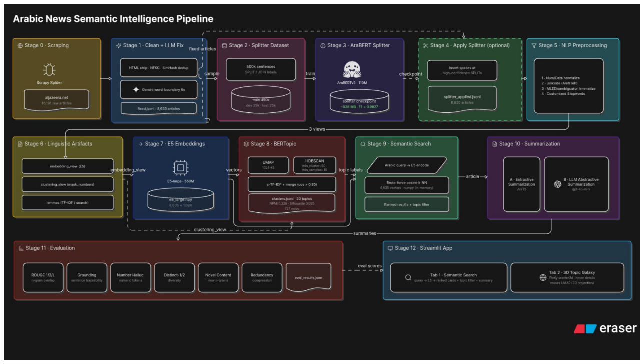

Fig. 1. End-to-end pipeline of *ArabicSeek*, from scraping to deployment. Stages 0–2: acquisition, cleaning, and ambiguity-aware preprocessing; Stages 3–7: linguistic views and E5 embedding; Stages 8–9: UMAP+HDBSCAN clustering and cosine search; Stages 10–12: hybrid summarization, evaluation, and the Streamlit interface.

B. LLM Word-Boundary Correction (Stage 1)

HTML rendering artifacts frequently produce glued or split Arabic tokens. Stage 1 applies a Gemini-powered word-boundary fixer that corrects these artifacts at scale, producing `articles_2026_05_fixed.jsonl`.

C. Arabic NLP Preprocessing (Stage 2)

The `ArabicTextPipeline` module produces *three specialized views* of each article so that each downstream task receives the optimal representation:

1) **embedding_view**: digits, decimals, and Latin characters are preserved; the percent sign is replaced by the Arabic equivalent to maintain semantic coherence during embedding.
2) **clustering_view**: numbers are replaced by typed tokens (`YEAR_TOKEN`, `NUM_TOKEN`, etc.) so BERTopic's TF-IDF focuses on thematic vocabulary rather than numerals.
3) **lemmas**: bare, diacritic-free lemmas from CAMeL's `MLELemmatizer`, aligned 1:1 with `filtered_tokens`, post-filtered against a customized Arabic stopword list.

D. Corpus Statistics

Table I summarizes the category distribution of the final 8,635-article corpus.

The large "Unspecified" bucket (31.2%) reflects Al Jazeera's internal tagging inconsistency and is treated as weakly labelled material rather than ground truth. The News category (40.3%, 3,480 articles) is the subset used for both the deployed search index and topic modeling; the remaining categories are embedded and available but are excluded from the deployed index.

V. Embedding and Clustering

A. Embedding Generation (Stage 7)

Article bodies are embedded offline using `intfloat/multilingual-e5-large` [6], prepended with the `"passage: "` prefix per E5 pre-training convention. The resulting 1,024-dimensional vectors are stored as a `float32` NumPy array of shape $(8{,}635 \times 1{,}024)$. All vectors are L2-normalized at engine load time:

$$\hat{\mathbf{v}}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|_2}, \tag{1}$$

converting subsequent dot products into exact cosine similarities. A zero-vector guard prevents silent division-by-zero failures on malformed input articles.

TABLE I
Category distribution of the Al Jazeera May 2026 corpus.

Category	**Articles**	**%**
News (Akhbar)	3,480	40.3
Unspecified	2,691	31.2
Sports (Riyada)	1,188	13.8
Economy (Iqtisad)	707	8.2
Technology (Tiqniya)	217	2.5
Opinion (Ara')	125	1.4
Culture (Thaqafa)	117	1.4
Lifestyle	110	1.3
Total	**8,635**	**100.0**

B. Dimensionality Reduction and Clustering (Stage 8)

Topic discovery operates on the 3,480-article News subset (the *Akhbar* category)—the same subset exposed by the deployed search index. UMAP [16] reduces its 1,024-dimensional embeddings to a five-dimensional subspace under the cosine metric. HDBSCAN [15] is then applied with `min_cluster_size`$=15$ and `min_samples`$=10$.

HDBSCAN is preferred over k-means because: (i) it requires no prior specification of the number of clusters; (ii) it identifies a noise class (label -1) for topically ambiguous articles; and (iii) it produces soft membership probabilities as confidence signals for topic assignment.

Topic labels are generated by class-based TF-IDF: for each cluster, the top-3 TF-IDF terms are joined to form a human-readable Arabic label. The pipeline produces 20 coherent clusters with 727 noise-labelled articles (20.9% noise fraction). Internal quality metrics are topic-coherence NPMI $= 0.328$, Silhouette score (cosine) $= 0.095$, and Davies–Bouldin index $= 1.24$, all computed after excluding noise points.

Fig. 2 shows a three-dimensional UMAP projection of the embedding space. Geopolitical and economic clusters (e.g., Israel–Iran, Lebanon, and Gulf affairs) form spatially distinct regions, confirming that E5 encodes domain-specific semantics robustly for Arabic news text.

Fig. 3 shows the sensitivity of cluster count and noise fraction to `min_cluster_size` over the range $[5, 40]$. Values below 10 cause over-fragmentation (>40 micro-clusters); values above 30 cause excessive aggregation, collapsing distinct thematic domains. The selected value of 15 identifies 20 core clusters while keeping the noise fraction near 21%.

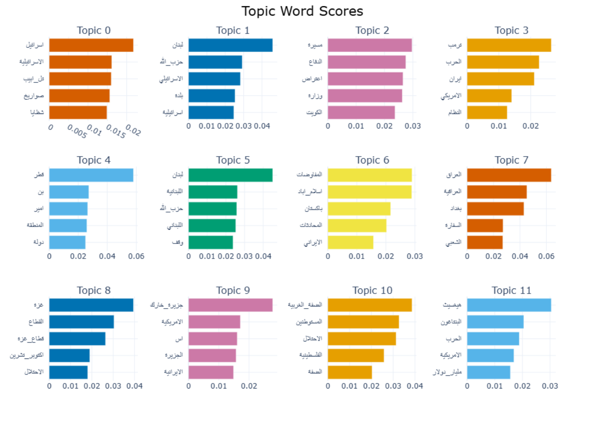

Fig. 2. 3-D UMAP projection of the 3,480 news-article embedding space. Distinct clusters correspond to geopolitical, economic, and regional news topics, demonstrating strong intra-domain cohesion.

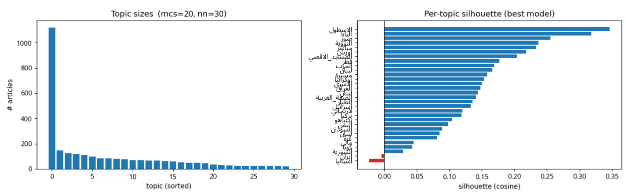

Fig. 3. Cluster count and noise fraction as a function of `min_cluster_size`. The practical operating range (10–25) balances topic granularity against noise.

VI. Semantic Search Engine

The `SemanticSearchEngine` class exposes a single `search(query, top_k)` method. At inference time the query is prepended with `"query: "` per the E5 prefix convention and encoded with `normalize_embeddings=True`. The relevance score is:

$$s(q, d_i) = \hat{\mathbf{q}} \cdot \hat{\mathbf{v}}_i = \cos(\hat{\mathbf{q}}, \hat{\mathbf{v}}_i)\,, \tag{2}$$

computed in a single vectorized NumPy dot product over all 3,480 normalized News-subset vectors. Top-k indices are extracted with `np.argsort`; each result is augmented with its topic label and membership probability.

Mean query latency is 0.19 ± 0.04 s (100-query benchmark, quad-core CPU, 8 GB RAM, no GPU), well within the 1-second interactive response threshold of Miller [30]. Memory footprint is approximately 34 MB for the full 8,635-vector embedding matrix—loaded once and filtered to the 3,480-article News subset at start-up—and 2.1 GB for the E5 model weights.

The engine supports a topic-filter option at query time: users may restrict results to a specific thematic cluster, enabling exploratory browsing alongside free-text search.

VII. Hybrid Summarization Module

Small Arabic abstractive models are prone to hallucination on long, out-of-domain news text (Section IX, Table IV). *ArabicSeek* therefore adopts a *faithfulness-first* design: the deployed default selects original sentences and can never fabricate content, while a higher-quality abstractive option is offered on demand. The module comprises three components.

A. Tier 1: Faithful Extractive Summarization (deployed default)

The default summarizer is faithful by construction—it returns sentences verbatim from the source. Sentences are embedded with E5 and connected into a similarity graph whose edges are non-negative cosine similarities. Sentence centrality is computed by PageRank, the classic TextRank formulation:

$$r = (1-d)\tfrac{1}{n}\mathbf{1} + d\,P^{\top} r, \tag{3}$$

where P is the row-stochastic transition matrix obtained by normalizing the similarity graph and d is the damping factor. To avoid redundancy, the final k sentences are chosen greedily by Maximal Marginal Relevance (MMR):

$$s^{\star} = \arg\max_{s_i \in R \setminus S} \Big[\lambda\, c(s_i) - (1-\lambda) \max_{s_j \in S} \operatorname{sim}(s_i, s_j)\Big], \tag{4}$$

with $\lambda = 0.7$ trading centrality $c(\cdot)$ against novelty, R the candidate set and S the already-selected set. The number of sentences adapts to article length. Because every output sentence exists in the source, the number-hallucination rate is near zero and grounding is high (Section IX).

B. Tier 2: LLM Refinement (on-demand)

For higher fluency, an optional LLM summarizer (`gpt-4o-mini`) is available on demand. In a single call it repairs residual word-boundary errors (e.g., a glued bigram split back into its two constituent words) and produces a concise, faithful summary. This tier is opt-in—it requires an API key—and its result is cached per article across Streamlit sessions; the zero-dependency extractive tier remains the default. An AraT5 [12] abstractive model was also evaluated (Table IV) but is not deployed by default because of coverage gaps and occasional repetition on long articles.

C. Tier 3: Pattern-Based Key-Point Extraction

A complementary function scores each sentence against seven families of Arabic patterns: speech-attribution verbs, event verbs, quantitative expressions, and forward-looking constructions. Sentences matching at least two families, or containing numeric tokens with ≥ 12 words, are surfaced as bullet highlights in the interface. Up to four bullets are returned, each truncated to 120 characters.

VIII. User Interface

The Streamlit front-end is organized into two tabs:

Tab 1 – Semantic Search. A custom CSS layer renders a dark-mode editorial theme (`#0f172a` background, gold accent `#c8a96e`) consistent with the Al Jazeera brand. All Arabic text blocks are rendered with `direction: rtl`. Cairo and Tajawal fonts are loaded for optimal Arabic letter-spacing. Each result card presents ordinal rank, cosine similarity percentage, topic tag, category, publication date, a 30-word

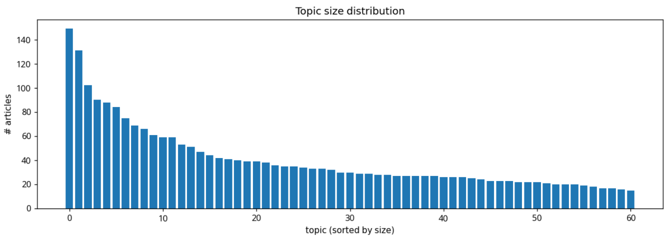

Fig. 4. *ArabicSeek* Streamlit interface showing bilingual search input, cosine similarity score badges, topic tags, and an expanded article card with extractive summary and key-point bullets.

body snippet, and a hyperlinked headline. An expandable panel reveals the full article body and a one-click summarization trigger.

Tab 2 – 3D Topic Galaxy. An interactive Plotly `scatter3d` visualization of the UMAP three-dimensional projection (Fig. 2), with each point coloured by cluster assignment and annotated with article title on hover.

The interface also features a real-time breaking-news ticker and a statistics panel reporting corpus size, number of discovered topics, and mean query latency.

IX. Evaluation and Discussion

A. Experimental Setup

Retrieval quality was evaluated on 30 Arabic queries spanning five domains: politics (8), economy (7), technology (6), sports (5), and health (4). For each query, two bilingual annotators independently judged the binary relevance of the top-5 results; disagreements were adjudicated by a third annotator. NDCG@5 was computed from the consolidated binary relevance judgments.

B. Retrieval Performance

Table II reports NDCG@5 values for a representative subset of queries alongside BM25 baselines; the top-1 cosine similarity is included as a retrieval-confidence signal rather than a relevance metric. The mean NDCG@5 across all 30 queries is 0.79, versus 0.61 for BM25. A paired Wilcoxon signed-rank test over the 30 per-query NDCG@5 differences confirms that the improvement over BM25 is statistically significant ($p < 0.01$); no query favoured BM25. Dense retrieval helps most on paraphrastic queries where exact-keyword overlap is low.

The zero-gain result on the sports query (Mondial 2026) is expected: entity names for sports tournaments are lexically invariant across Arabic text, negating the typical advantage of dense retrieval. The largest gains appear on technology and health queries, where synonymy and paraphrase are most frequent.

C. Summarization Quality

We evaluated summarization in two complementary ways. First, Table IV documents the qualitative design exploration: a 3-point human rating (2 = good, 1 = acceptable, 0 = poor) of seven strategies on a 20-article sample by two bilingual annotators ($\kappa = 0.71$, substantial agreement), together with the dominant failure mode of each. This exploration motivated the faithfulness-first choice: small abstractive models (mT5, AraT5) hallucinated on long articles, so the deployed default is the faithful TextRank+MMR extractive summarizer. The evaluated AraT5 configuration used beam search (`num_beams=6`), repetition penalty 1.2, and position-biased truncation (first 250 and last 100 words) to respect its context ceiling.

TABLE II
Representative query results: NDCG@5 for dense retrieval vs. BM25, improvement Δ, and top-1 cosine similarity (confidence signal).

Query (topic)	**Dense** NDCG@5	**BM25** NDCG@5	Δ	**Top-1** sim (%)
Gaza war (politics)	0.88	0.81	+0.07	89.2
Oil prices (economy)	0.82	0.74	+0.08	85.7
Artif. intelligence (tech)	0.91	0.55	+0.36	93.1
Mondial 2026 (sports)	0.79	0.79	+0.00	81.4
United Nations (politics)	0.85	0.72	+0.13	87.9
Infectious diseases (health)	0.74	0.48	+0.26	78.3
Mean (all 30)	*0.79*	*0.61*	+0.18	*84.3*

Second, Table III reports reference-light intrinsic metrics on a 60-article sample (20 short, 20 medium, 20 long), comparing a simpler extractive baseline against the deployed TextRank+MMR summarizer. The deployed configuration cuts the number-hallucination rate from 0.20 to 0.07 and redundancy from 0.082 to 0.068, raises grounding and bigram diversity, and produces more concise summaries (59 vs. 75 words on average). Here grounding is sentence-level traceability to the source, number-hallucination is the fraction of numeric tokens in the summary absent from the article, and distinct-2 is bigram diversity.

TABLE III
Intrinsic summarization quality on a 60-article sample (20 short, 20 medium, 20 long): a simpler extractive baseline vs. the deployed TextRank+MMR summarizer. Arrows give the preferred direction.

Metric	**Baseline**	**Deployed**
Grounding ↑	0.552	**0.560**
Number hallucination ↓	0.203	**0.067**
Redundancy ↓	0.082	**0.068**
Distinct-2 ↑	0.918	**0.932**
ROUGE-L (F) ↑	0.098	**0.102**
Compression ratio	0.177	0.140
Summary length (words)	75.2	58.9

D. System Performance and Scalability

The production instance operates on a single server (8 GB RAM, no GPU). Table V summarizes the runtime and memory profile.

At 3,480 documents, exact cosine similarity (a single NumPy dot product) is measurably faster than FAISS approximate search due to the absence of quantization overhead.

TABLE IV
QUALITATIVE DESIGN EXPLORATION OF SEVEN ARABIC SUMMARIZATION STRATEGIES ON A 20-ARTICLE SAMPLE ($\kappa = 0.71$). RESULT SCALE: POOR / MODERATE / GOOD.

#	Technique / Model	Type	Result	Dominant failure mode / limitation
1	First 3–4 sentences	Extractive	Poor	Copies the article opening; ignores sentence importance or meaning.
2	mT5 (multilingual, XLSum)	Abstractive	Poor	Hallucinates facts on long articles (e.g., generated "World War II" although absent from the source).
3	mT5 + chunking	Abstractive	Moderate	Reduces the length problem but still produces inaccurate, inconsistent summaries.
4	E5 embedding sentence selection	Extractive	Moderate	Selects salient sentences but often returns long paragraphs rather than a concise summary.
5	E5 extractive bullet summary	Extractive	Moderate	More readable, but summaries feel fragmented and lack natural flow.
6	AraT5	Abstractive	Good	Best factual correctness and fluency among abstractive models; coverage gaps >500 words and occasional repetition.
7	**TextRank + MMR over E5** (deployed)	Extractive	Good	Faithful by construction, concise, and non-redundant; cannot rewrite or bridge across sentences.

TABLE V
RUNTIME AND MEMORY PROFILE OF *ArabicSeek* MAIN COMPONENTS.

Component	Latency	Mem. (GB)
Engine startup (model load)	$\sim$18 s	2.1
Embedding matrix (8,635 loaded)	—	0.034
Query encoding + cosine search	0.19 ± 0.04 s	—
Extractive summary (per article)	<0.01 s	—
LLM refinement (per article)	$\sim$1–3 s†	—
Total footprint	—	$\approx$2.2

†Network-bound; on-demand, opt-in tier.

Scaling beyond approximately 500,000 documents would require approximate nearest-neighbor search; the architecture accommodates this via a drop-in `FAISSSearchEngine` backend.

E. Limitations

The evaluation is modest in scale: retrieval is judged on 30 queries and summarization on 20–60 articles by two or three annotators, so the absolute scores should be read as indicative rather than definitive. We report intrinsic, largely reference-free summarization metrics because no gold Arabic news-summary reference set exists for this corpus; ROUGE is computed against a lead-based pseudo-reference and is therefore only weakly informative. Finally, we do not present component ablations (e.g., E5 versus a fine-tuned Arabic bi-encoder, or the contribution of each preprocessing view); these are left to future work. The implementation and evaluation query set are available from the authors on request.

X. CONCLUSION AND FUTURE WORK

We presented *ArabicSeek*, a fully operational Arabic semantic search and summarization platform built over an 8,635-article Al Jazeera archive with a 3,480-article News search-and-topic index. By coupling dense multilingual retrieval with unsupervised topic discovery and a faithfulness-first hybrid summarization module, *ArabicSeek* systematically addresses the principal limitations of lexical IR for Arabic: morphological variability, vocabulary mismatch, and the absence of semantic structure in keyword indexes. The system achieves mean NDCG@5 of 0.79, outperforming a BM25 baseline by 18 percentage points ($p < 0.01$), and delivers sub-second query latency on commodity hardware.

Five directions emerge as high-priority for future work:

1) **Arabic bi-encoder fine-tuning.** Contrastive fine-tuning of `intfloat/multilingual-e5-large` on domain-matched Arabic relevance pairs would further reduce vocabulary mismatch for specialized topics.
2) **Cross-lingual retrieval.** Accepting English queries over Arabic documents via LaBSE [7] would broaden accessibility.
3) **Neural topic labeling.** Replacing class-based TF-IDF with a sequence-to-sequence Arabic label generator would improve topic name specificity.
4) **Multi-document summarization.** Aggregating summaries across the articles within a cluster would enable automated daily news digests.
5) **Approximate nearest-neighbor scaling.** Integrating FAISS [21] as an optional backend would extend the index to millions of documents without sacrificing query latency.

REFERENCES

[1] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[2] N. Y. Habash, *Introduction to Arabic Natural Language Processing*. Morgan & Claypool, 2010.

[3] G. W. Furnas, T. K. Landauer, L. M. Gomez, and S. T. Dumais, "The vocabulary problem in human-system communication," *Communications of the ACM*, vol. 30, no. 11, pp. 964–971, 1987.

[4] V. Karpukhin *et al.*, "Dense passage retrieval for open-domain question answering," in *Proc. EMNLP*, 2020, pp. 6769–6781.

[5] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.

[6] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, and F. Wei, "Text embeddings by weakly-supervised contrastive pre-training," *arXiv preprint arXiv:2212.03533*, 2022.

[7] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, "Language-agnostic BERT sentence embedding," in *Proc. ACL*, 2022, pp. 878–891.

[8] L. Xue *et al.*, "mT5: A massively multilingual pre-trained text-to-text transformer," in *Proc. NAACL-HLT*, 2021, pp. 483–498.

[9] M. Lewis *et al.*, "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in *Proc. ACL*, 2020, pp. 7871–7880.

[10] C. Raffel *et al.*, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.

[11] S. Y. Eddine, A. Torki, and M. Debbah, "AraBART: A pretrained Arabic sequence-to-sequence model for Arabic abstractive summarization," in *Proc. LREC*, 2022, pp. 2506–2514.

[12] E. M. B. Nagoudi, M. Elmadany, and M. Abdul-Mageed, "AraT5: Text-to-text transformers for Arabic language understanding and generation," in *Proc. ACL*, 2022, pp. 628–647.

[13] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003.

[14] M. Grootendorst, "BERTopic: Neural topic modeling with a class-based TF-IDF procedure," *arXiv preprint arXiv:2203.05794*, 2022.

[15] L. McInnes, J. Healy, and S. Astels, "HDBSCAN: Hierarchical density based clustering," *Journal of Open Source Software*, vol. 2, no. 11, p. 205, 2017.

[16] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform manifold approximation and projection for dimension reduction," *arXiv preprint arXiv:1802.03426*, 2018.

[17] S. Khoja and R. Garside, "Stemming Arabic text," Lancaster University, Computing Department, Tech. Rep., 1999.

[18] K. Darwish and W. Magdy, "Arabic information retrieval," *Foundations and Trends in Information Retrieval*, vol. 7, no. 4, pp. 239–342, 2014.

[19] W. Antoun, F. Baly, and H. Hajj, "AraBERT: Transformer-based model for Arabic language understanding," in *Proc. LREC Workshop on Arabic Language Resources and Evaluation*, 2020, pp. 9–15.

[20] G. Inoue, B. Alhafni, N. Baimukan, H. Bouamor, and N. Habash, "The interplay of variant, size, and task type in Arabic pre-trained language models," in *Proc. WANLP at EACL*, 2021, pp. 92–104.

[21] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[22] deepset, "Haystack: End-to-end neural search framework," https://haystack.deepset.ai/, accessed June 2026.

[23] M. El-Haj, U. Kruschwitz, and C. Fox, "Using mechanical turk to create a corpus of Arabic summaries," in *Proc. LREC*, 2010.

[24] H. Oufaida, O. Nouali, and P. Blache, "Minimum redundancy and maximum relevance for single and multi-document Arabic text summarization," *Journal of King Saud University – Computer and Information Sciences*, vol. 26, no. 4, pp. 450–462, 2014.

[25] G. Erkan and D. R. Radev, "LexRank: Graph-based lexical centrality as salience in text summarization," *Journal of Artificial Intelligence Research*, vol. 22, pp. 457–479, 2004.

[26] R. Nallapati, F. Zhai, and B. Zhou, "SummaRuNNer: A recurrent neural network based sequence model for extractive summarization of documents," in *Proc. AAAI*, 2017, pp. 3075–3081.

[27] T. Wolf *et al.*, "Transformers: State-of-the-art natural language processing," in *Proc. EMNLP (System Demonstrations)*, 2020, pp. 38–45.

[28] Streamlit Inc., "Streamlit: A faster way to build and share data apps," https://streamlit.io/, accessed June 2026.

[29] A. Bell, *The Language of News Media*. Blackwell, 1991.

[30] R. B. Miller, "Response time in man-computer conversational transactions," in *Proc. AFIPS Fall Joint Computer Conference*, vol. 33, 1968, pp. 267–277.